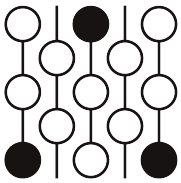
Review of Problem 8 – Astra

First Proof Third Batch
Reviewer 1
October 8–9, 2026



1 Review

Correctness.

- 1: essentially flawless mathematical argument
- 0.5: minor errors that are easily corrected
- 0: blank solution, or acknowledged partial result
- −0.5: major errors
- −1: proposed proof makes no progress towards the result.

Acknowledges that it doesn't solve problem.
Score: 0

Exposition.

- 1: well-written and clear solution.
- 0: argument that is possible to follow with serious effort
- −1: very poor exposition

Score:0

Attributions.

- 1: solutions contains fully appropriate attributions to related work and theorems being used.
- 0: moderate issues with missing attributions or hallucinated references
- −1: blatant plagiarism or hallucinations

Score: 0

> **Reviewer 1**
>
> The implication from flow to diffeomorphism has been known for ever though suspension, it is a simple exercise and it is already mentioned in a paper of the reviewer with coauthor, so there is not much value in this step. Even the other direction would represent some interest, though quite unrelated with the main problem. Exposition is not bad, easy reading but mostly because mathematics is trivial.

2 Submission with inline comments

Abstract

A complete proof of the proposed assertion is not known in the stated generality. In fact, the assertion would settle the open problem of vanishing Lyapunov exponents for cohomologically rigid diffeomorphisms. We give a complete proof of this reduction, explaining why the usual structural results about globally hypoelliptic vector fields do not supply the requested conclusion.

3 Status of the question

The hypothesis in the question is called *global hypoellipticity*. The established theory relates this property to cohomology freeness and gives unique ergodicity with a smooth invariant probability density; see Forni [1, Proposition 2.4 and Theorem 3.10]. It does not, however, supply the proposed general entropy conclusion.

The relevant unresolved issue is explicitly recorded by Karaliolios [3, Introduction, p. 6]: it is unknown whether cohomological rigidity of a diffeomorphism implies that all its Lyapunov exponents vanish. The suspension construction below shows that a proof of the assertion in the question would imply precisely this vanishing result.

Thus the following is a rigorous reduction to an open problem, rather than a purported proof of the requested assertion.

4 A sufficient condition for global hypoellipticity

We use smooth positive densities, so that orientability is unnecessary. If μ is such a density, scalar distributions are identified with continuous linear functionals on $C^\infty(M)$ through the pairing

$$\langle u, \phi\rangle_\mu = \langle u, \phi\mu\rangle.$$

For a vector field preserving μ, the distributional derivative satisfies

$$\langle Xu, \phi\rangle_\mu = -\langle u, X\phi\rangle_\mu.$$

Lemma 1. *Let Y be a smooth vector field on a compact manifold P, preserving a smooth positive probability density μ. Suppose that for every $f \in C^\infty(P)$ there is $v \in C^\infty(P)$ such that*

$$Yv = f - \int_P f\, d\mu. \tag{1}$$

Then Y satisfies the global hypoellipticity hypothesis of the question.

Proof. Suppose that $Yu = f$ for $u \in \mathcal{D}'(P)$ and $f \in C^\infty(P)$. Invariance of μ gives

$$\int_P f\, d\mu = \langle Yu, 1\rangle_\mu = -\langle u, Y1\rangle_\mu = 0.$$

Consequently, (1) supplies a smooth function v with $Yv = f$. Set $w = u - v$, so that $Yw = 0$.

For any $\phi \in C^\infty(P)$, choose $a \in C^\infty(P)$ satisfying

$$Ya = \phi - \int_P \phi\, d\mu.$$

Then

$$\langle w, \phi\rangle_\mu = \left(\int_P \phi\, d\mu\right)\langle w, 1\rangle_\mu + \langle w, Ya\rangle_\mu = \left(\int_P \phi\, d\mu\right)\langle w, 1\rangle_\mu.$$

Thus w is a constant distribution. It follows that $u = v + w$ is smooth. □

5 Suspending a cohomologically rigid diffeomorphism

Definition 2. Let $F : N \to N$ be a smooth diffeomorphism of a compact connected manifold, preserving a smooth positive probability density ν. We call F *cohomologically rigid* if, for every $a \in C^\infty(N)$ with $\int_N a \, d\nu = 0$, there is $v \in C^\infty(N)$ satisfying

$$v \circ F - v = a. \tag{2}$$

Lemma 3. *The constant-roof suspension of a cohomologically rigid diffeomorphism has a globally hypoelliptic generating vector field.*

Proof. Let

$$P_F = (N \times \mathbb{R})/\!\sim, \qquad (x, s+1) \sim (Fx, s).$$

The equivalence relation is generated by the free, properly discontinuous smooth $\mathbb{Z}$-action

$$k \cdot (x, s) = (F^k x, s - k).$$

Hence P_F is a smooth manifold. It is compact because the image of $N \times [0,1]$ is all of P_F, and it is connected because N is connected.

Translation in the second coordinate descends to the smooth flow

$$G_t[x, s] = [x, s + t],$$

whose generating vector field Y is induced by ∂_s. Since F preserves ν, the product density $d\nu \, ds$ descends to a smooth positive G_t-invariant probability density μ on P_F.

We verify (1). Given $f \in C^\infty(P_F)$, denote its lift to $N \times \mathbb{R}$ by $\widetilde{f}$. Thus

$$\widetilde{f}(x, s+1) = \widetilde{f}(Fx, s).$$

Set

$$c = \int_{P_F} f \, d\mu, \qquad a(x) = \int_0^1 \big(\widetilde{f}(x, r) - c\big) \, dr.$$

By the definition of the suspension measure, $\int_N a \, d\nu = 0$. Cohomological rigidity supplies $v \in C^\infty(N)$ with $v \circ F - v = a$.

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$$\frac{\partial}{\partial s}\big(\widetilde{u}(x, s+1) - \widetilde{u}(Fx, s)\big) = 0,$$

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$$Yu = f - c = f - \int_{P_F} f \, d\mu.$$

Lemma 1 now proves global hypoellipticity. □

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Proposition 4. *If the assertion in the question holds for all compact connected smooth manifolds, then every cohomologically rigid smooth diffeomorphism preserving a smooth positive probability density has zero topological entropy and all its Lyapunov exponents vanish almost everywhere.*

Proof. Let $F : N \to N$ be such a diffeomorphism and form the suspension from Lemma 3. Its generating vector field satisfies exactly the hypothesis of the question.

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$$h_{\mathrm{top}}(F) \le h_{\mathrm{top}}(G_1). \tag{3}$$

The entropy of a flow is, by convention, that of its time-one map. Consequently, the requested assertion would give $h_{\mathrm{top}}(G_1) = 0$, and (3) would imply $h_{\mathrm{top}}(F) = 0$.

Metric entropy is bounded above by topological entropy [2], so

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The integrand is nonnegative. Its integral being zero, every Lyapunov exponent is nonpositive at ν-almost every point.

Finally, choose a Riemannian volume density m and write $d\nu = \rho\, dm$, where ρ is smooth and everywhere positive. Invariance of ν gives

$$\left|\det D_x F^n\right| = \frac{\rho(x)}{\rho(F^n x)}.$$

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$$\sum_{j=1}^{d} \lambda_j(x) = \lim_{n\to\infty} \frac{1}{n} \log\left|\det D_x F^n\right| = 0$$

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Remark (The essential gap in a tempting entropy argument). Preservation of a smooth volume gives $\sum_j \lambda_j = 0$, not $\sum_j \max\{\lambda_j, 0\} = 0$. The latter is the quantity occurring in Pesin's entropy formula. Consequently, replacing entropy by the integral of the logarithm of the full Jacobian determinant would be incorrect.

7 Conclusion

Proposition 4 shows that the question includes the Lyapunov-exponent vanishing problem explicitly identified as open in [3]. The established global-hypoellipticity results provide cohomological rigidity and a smooth invariant measure, but not the additional vanishing needed for entropy. Therefore a complete proof of the proposed general assertion cannot presently be supplied as an established mathematical result; doing so would require resolving this open problem.

References

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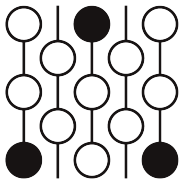
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Review of Problem 8 – Astra

First Proof Third Batch
Reviewer 2

October 8–9, 2026



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- -0.5: major errors
- -1: proposed proof makes no progress towards the result.

Note that points should be docked for wrong statements/ solutions.
Score: 0. The argument involves a trivial manipulation that doesn't solve the problem, or bring anyone closer to the solution.

Exposition.

- 1: well-written and clear solution.
- 0: argument that is possible to follow with serious effort
- -1: very poor exposition

Score: 0. Not much of a solution.

Attributions.

- 1: solutions contains fully appropriate attributions to related work and theorems being used.
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Score: 0. The reduction from continuous to discrete is in the literature.

2 Submission with inline comments

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> **Reviewer 2**
>
> The reduction of the discrete time to continuous time version is available in the paper of Rodriguez-Hertz–Rodriguez-Hertz, and overall the reductions in this text are straightforward and are not much progress. On the positive side, the paper is quite explicit that it doesn't solve the problem.

Proof. Let $F : N \to N$ be such a diffeomorphism and form the suspension from Lemma 3. Its generating vector field satisfies exactly the hypothesis of the question.

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Remark (The essential gap in a tempting entropy argument). Preservation of a smooth volume gives $\sum_j \lambda_j = 0$, not $\sum_j \max\{\lambda_j, 0\} = 0$. The latter is the quantity occurring in Pesin's entropy formula. Consequently, replacing entropy by the integral of the logarithm of the full Jacobian determinant would be incorrect.

7 Conclusion

Proposition 4 shows that the question includes the Lyapunov-exponent vanishing problem explicitly identified as open in [3]. The established global-hypoellipticity results provide cohomological rigidity and a smooth invariant measure, but not the additional vanishing needed for entropy. Therefore a complete proof of the proposed general assertion cannot presently be supplied as an established mathematical result; doing so would require resolving this open problem.

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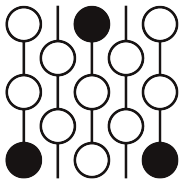
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Review of Problem 8 – Astra

First Proof Third Batch
Reviewer 3
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1 Review

Wren correctly notes that it does not prove the problem statement, and instead provides some discussion in and around the conjecture: Section 4 relates global hypoellipticity to solubility of the related cohomological equation (1); and Sections 5 and 6 relate global hypoellipticity for flows to an analogous property for diffeomorphisms.

Essentially nothing new is proved here. The statements are elementary and could be read off of the discussion in [1] and some additional brief computations. Accurate but trivial. For what it is worth, the exposition of these trivial remarks is relatively good, hence my score here.

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Score: 0

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Metric entropy is bounded above by topological entropy [2], so

$$h_\nu(F) = 0.$$

Let $\lambda_1(x), \dots, \lambda_d(x)$ denote the Lyapunov exponents, listed with multiplicity. Since F is smooth and ν is a smooth invariant probability density, Pesin's entropy formula applies [4]:

$$h_\nu(F) = \int_N \sum_{j=1}^{d} \max\{\lambda_j(x), 0\}\, d\nu(x).$$

The integrand is nonnegative. Its integral being zero, every Lyapunov exponent is nonpositive at ν-almost every point.

Finally, choose a Riemannian volume density m and write $d\nu = \rho\, dm$, where ρ is smooth and everywhere positive. Invariance of ν gives

$$\left|\det D_x F^n\right| = \frac{\rho(x)}{\rho(F^n x)}.$$

Because ρ is bounded above and bounded away from zero,

$$\sum_{j=1}^{d} \lambda_j(x) = \lim_{n\to\infty} \frac{1}{n} \log\left|\det D_x F^n\right| = 0$$

at every Lyapunov-regular point. A finite collection of nonpositive numbers with sum zero consists entirely of zeros. Thus all Lyapunov exponents vanish ν-almost everywhere. □

Remark (The essential gap in a tempting entropy argument). Preservation of a smooth volume gives $\sum_j \lambda_j = 0$, not $\sum_j \max\{\lambda_j, 0\} = 0$. The latter is the quantity occurring in Pesin's entropy formula. Consequently, replacing entropy by the integral of the logarithm of the full Jacobian determinant would be incorrect.

7 Conclusion

Proposition 4 shows that the question includes the Lyapunov-exponent vanishing problem explicitly identified as open in [3]. The established global-hypoellipticity results provide cohomological rigidity and a smooth invariant measure, but not the additional vanishing needed for entropy. Therefore a complete proof of the proposed general assertion cannot presently be supplied as an established mathematical result; doing so would require resolving this open problem.

References

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